

STAGE 1 — Feature extraction

Xception, VGG16/19, ResNet, Inception, MobileNet, DenseNet, EfficientNet (25 CNN extractors)

STAGE 2 — Dimensionality reduction / selection

PCA

UMAP

Relief-F

Full

Selected feature vector — 2, 10, ..., 300 dims (or full)

STAGE 3 — Classification

SVM (linear) · SVM (RBF) · MLP · Logistic Regression · Random Forest · AdaBoost · Decision Tree · Naïve Bayes

Clear path

Obstructed path

LOOCV accuracy: 0.9474 • 10-fold accuracy: 0.9412